Evaluation

I. Choose the best answer.

1. The correct statement regarding the comparison of staggered and eclipsed conformations of ethane, is (NEET)

   a) the eclipsed conformation of ethane is more stable than staggered conformation even though the eclipsed conformation has torsional strain.

   b) the staggered conformation of ethane is more stable than eclipsed conformation, because staggered conformation has no torsional strain.

   c) the staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has torsional strain.

   d) the staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has no torsional strain.

2. $C_2H_5Br + 2Na \xrightarrow{\text{dry ether}} C_4H_{10} + 2NaBr$ The above reaction is an example of which of the following

   a) Reimer Tiemann reaction

   b) Wurtz reaction

   c) Aldol condensation

   d) Hoffmann reaction

3. An alkyl bromide (A) reacts with sodium in ether to form 4, 5– diethyloctane, the compound (A) is

   a) $CH_3(CH_2)_3Br$

   b) $CH_3(CH_2)_5Br$

   c) $CH_3(CH_2)_3CH(Br)CH_3$

   d) $CH_3 - (CH_2)_2 - CH(Br) - CH_2 - CH_3$

4. The C – H bond and C – C bond in ethane are formed by which of the following types of overlap

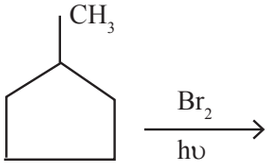
   a) $sp^3 - s$ and $sp^3 - sp^3$

   b) $sp^2 - s$ and $sp^2 - Sp^2$

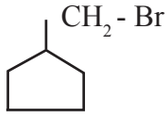
   c) sp – sp and sp – sp

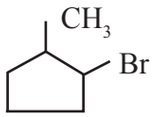
   d) p – s and p – p

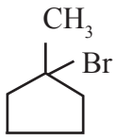
5. In the following reaction,

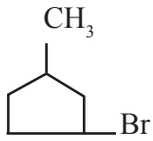


The major product obtained is

a) 

b) 

c) 

d) 

6. Which of the following is optically active

   a) 2 – methyl pentane

   b) citric acid

   c) Glycerol

   d) none of of these

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